CS-E4960 - Software Testing and Quality Assurance D, Lecture, 3.9.2024-26.11.2024

This course space end date is set to 26.11.2024 **Search Courses: CS-E4960**

Static Code Analysis 2: SonarQube

To do: Make a submission **To do:** Receive a grade

Opened: Tuesday, 22 October 2024, 10:15 AM
Due: Tuesday, 5 November 2024, 10:00 AM

Objective

In this assignment, your goal is to analyse the quality of a software project with the SonarQube tool.

Download and unzip the SonarQube server (Community edition): https://www.sonarsource.com/open-source-editions/sonarqube-community-edition/

To start the server, follow the instructions at: https://docs.sonarsource.com/sonarqube/latest/try-out-sonarqube/

Also, make sure that you have Java 17+ installed on your computer or VM.

Analyse the source code of the Oscar project using the SonarQube. Use the attached zip file, which contains the version of the Oscar source code, to analyse.

Detailed instructions for getting SonarQube set up are provided in the installation instructions file below.

Instructions

To complete this assignment, do the following:

1. Answer the following questions:
   - How would you interpret the overall quality of the project?
   - What are the most common issues, e.g. in terms of type, severity, and quality attribute?
   - Are there any particular parts of the project that seem more problematic regarding discovered issues?
   - Are the discovered issues relevant, or are there any false positives?
2. Modify the project source by injecting at least 3 new issues into the project. Use at least one issue per different issue type (Bug, Vulnerability, Code smell) and rerun the analysis. (Note: you can check supported issue detection rules for specific languages in the Rule menu of the Sonarqube server).
   - Briefly report on the following:
   - Which issues did you insert?
   - Were the issues discovered by the SonarQube?
   - If SonarQube would not be available, can you think of other ways these issues could be detected?

Submission

Submit a PDF file containing answers to the questions above. In the file, also include screenshots of the project analysis results in your SonarQube instance relevant to your answers, e.g. the view of the project dashboard, metrics, issues, etc. and the sou rce code view with your issues injected.

Grading:

Part 1: Successful analysis of the project with detailed interpretation of the analysis results concerning the specified questions (2p). The project was analysed successfully, but the result interpretation is shallow (1p). The project was not properly analyz ed, and the interpretation of results is incomplete or contains errors (0p)

Part 2: Three different types of issues were injected and their details are well explained in answers to the questions (2p). Three issues were injected, but they are not of different types, and/or the answers to questions are too shallow (1p). The issues w ere not injected properly, and the answers to the questions are incorrect (0p)

Use of AI and Collaboration with Other Students: Using AI tools to produce an answer for this assignment is not permitted.All parts of the submission must be produced by yourself (apart from files provided in the assignment). Collaboration with other students is not permitted. This is an individual assignment that you must author independently.

django-oscar-master.zip	18 October 2024, 2:02 AM
SonarQube Installation.pdf	18 October 2024, 2:02 AM

Submission status

Attempt number	This is attempt 1.
Submission status	No submissions have been made yet
Grading status	Not marked
Time remaining	Assignment is overdue by: 127 days 3 hours
Last modified	-